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Deep Transfer Learning for Multi-Crop Plant Disease Classification: A Cross-Dataset Benchmark with Class-Weighted, Attention-Enhanced ResNet50

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Data Availability Before publication, Authors are required to make fully available and without restriction all data underlying their findings. Sharing the data underlying your findings increases the visibility, credibility, and impact of your work, while supporting transparency and future scientific discovery. Please see our PLOS Data	No new data was generated by this study. All datasets analyzed during the current study are publicly available in online repositories as follows: 1. PlantVillage dataset: Available on Kaggle at https://www.kaggle.com/datasets/emmarex/plantdisease 2. Corn or Maize Leaf Disease dataset: Available on Kaggle at https://www.kaggle.com/datasets/smaranjitghose/corn-or-maize-leaf-disease-dataset 3. Mango Leaf Disease dataset (MangoLeafBD): Available on Kaggle at https://www.kaggle.com/datasets/aryashah2k/mango-leaf-disease-dataset All code and evaluation protocols necessary to reproduce the findings are described in

Policy page for detailed information on this policy. A **Data Availability Statement**, detailing where the data can be accessed, is required at first submission. Insert your Data Availability Statement in the box below. Please see the data reporting section of our submission guidelines for instructions on what you need to include in your Data Availability Statement. Examples of acceptable Data Availability Statements are included here for reference: 1. The minimal data set [and accompanying code, where generated] is available at [repository name] via [DOI hyperlink] 2. No data was generated by this study. The following existing data sources were used: [dataset name] from [dataset location] available via [web address or DOI hyperlink]. 3. For Study Protocols: No datasets were generated or analysed during the current study. All relevant data from this study will be made available upon study completion. *PLOS allows rare exemptions to address legal and ethical concerns. If you have legal or ethical restrictions, please use the box below to detail these in full sentences for the Journal team to consider.* This statement will be typeset if the manuscript is accepted for publication.	full within the manuscript.

Cover Letter

Nov 5, 2026

To
The Editor-in-Chief
PLOS ONE

Subject: *Submission of manuscript "Deep Transfer Learning for Multi-Crop Plant Disease Classification: A Cross-Dataset Benchmark with Class-Weighted, Attention-Enhanced ResNet50"*

Dear Editor-in-Chief,

We are submitting our original research manuscript, "Deep Transfer Learning for Multi-Crop Plant Disease Classification: A Cross-Dataset Benchmark with Class-Weighted, Attention-Enhanced ResNet50," for publication in PLOS ONE.

While deep learning approaches have shown impressive performance on plant leaf disease detection, most studies rely on isolated evaluations on single, uniform benchmarks. In this work, we conduct a cross-dataset study using ImageNet-pretrained ResNet50 across three public datasets with varied image conditions and class balances: PlantVillage (15 classes, 20,638 images; 21.11:1 imbalance ratio), Corn/Maize Leaf Disease (4 classes, 4,188 images; 2.28:1 ratio), and MangoLeafBD (8 classes, 4,000 images; balanced).

Across stratified five-fold cross-validation and a seven-factor ablation, we found that full-network fine-tuning is indispensable, outperforming frozen-backbone linear probing by 24.94% macro-F1 (0.9931 vs. 0.7437). We also evaluated two targeted enhancements: an inverse-frequency class-weighted cross-entropy loss and a terminal Squeeze-and-Excitation (SE) attention block, which adds only 2.23% parameters. Both techniques selectively improved macro-F1 on the more challenging and imbalanced sets (PlantVillage and Corn/Maize) without degrading the ceiling performance on Mango (1.0000 macro-F1). Their combination achieved a top macro-F1 of 0.9574 on Corn/Maize. Furthermore, the model is well-calibrated (ECE = 0.0149 on Corn/Maize), and Grad-CAM visualizations demonstrate that 45% of test errors between Grey Leaf Spot and Blight stem from authentic morphological lesion overlap rather than background artifacts.
This manuscript is original and is not under consideration elsewhere. All authors have reviewed and approved the submission. The authors declare that they have no competing interests. The datasets used are publicly available through the sources referenced in the text, and no external funding was received for this study.
Thank you for your time and consideration.

Sincerely,

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Deep Transfer Learning for Multi-Crop Plant Disease Classification: A Cross-Dataset Benchmark with Class-Weighted, Attention-Enhanced ResNet50

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Abstract

The early and accurate detection of plant leaf diseases helps to reduce crop losses and protect food security. This is especially important in places where manual inspection is still the main way diseases are diagnosed. We test transfer learning based on ResNet50 on three public heterogeneous datasets: PlantVillage (15 classes, 20,638 images), Corn/Maize leaf disease dataset (4 classes, 4,188 images) and Mango leaf disease dataset (8 classes, 4,000 images). A pretrained ResNet50 achieved 95.0-99.9% macro-F1 across the three datasets using stratified 5-fold cross-validation, with statistically significant differences between datasets (paired t-test, $p < 0.05$, large effect sizes). In a controlled ablation study, fine-tuning the full network outperforms training only a frozen classification head by a large margin (macro-F1 0.9931 vs. 0.7437) and a class-imbalance aware weighted loss is helpful on imbalanced datasets without hurting balanced ones. We test two lightweight, independently motivated improvements on this result: class-weighted fine-tuning and adding a Squeeze-and-Excitation (SE) attention block to ResNet50's final feature map (+2.2% parameters). Neither change affected the near-perfect Mango dataset, but both increased macro-F1 on the two datasets that had room for improvement (PlantVillage and Corn/Maize). This pattern is unlikely to be coincidental, since it follows the intended mechanism of both techniques. The combination of both techniques achieved the best performance on the most challenging dataset tested, Corn/Maize, with a macro-F1 of 0.9574. The resulting model is also well calibrated (Expected Calibration Error = 0.0149) and Gradient-weighted Class Activation Mapping (Grad-CAM) shows that its single most common mistake, Grey Leaf Spot read as Blight, comprising 45% of all test errors on Corn/Maize, can be attributed to true visual similarity between the two diseases, not an arbitrary failure mode. Taken together, these results make the case for class-weighted SE-attention-enhanced ResNet50 as an effective, well-calibrated, and interpretable recipe for plant disease classification across multiple crops.

Keywords: plant disease classification, transfer learning, ResNet50, convolutional neural networks, Squeeze-and-Excitation attention, class imbalance, cross-validation, explainable AI, Grad-CAM, precision agriculture

Introduction

Plant diseases destroy a large proportion of the world's crop production each year and smallholder farmers in developing regions suffer the most, as they have the least access to agricultural extension services and expert diagnosis. Trained agronomists can detect most disease early by visual inspection, but the method is slow and subjective and does not scale to the area under cultivation in many regions. Convolutional neural networks have somewhat changed this picture: transfer learning from large image datasets such as ImageNet has yielded strong plant disease classifiers and opened a path toward low-cost, smartphone-deployable diagnosis.

Most work to date, however, tests a single architecture on a single dataset, often the widely used PlantVillage set, collected in controlled laboratory conditions with uniform backgrounds. This raises two questions that motivate this study. First, does good performance on a controlled dataset transfer to other crops and disease types with different class counts, image conditions, and class-balance characteristics? Second, putting raw accuracy aside, how robust, well calibrated and interpretable are these models — properties that matter for real deployment, but are rarely reported alongside the headline number?

We answer both questions with a systematic multi-dataset evaluation of ResNet50 for the task of plant disease classification. The contributions of this work are

- A controlled benchmark of ResNet50 on three heterogeneous publicly available plant disease datasets (PlantVillage, Corn/Maize, Mango) run under stratified 5-fold cross-validation with 95% confidence intervals and pairwise statistical significance testing.
- A seven-factor ablation study to isolate the individual contribution of pretraining, data augmentation, fine-tuning strategy, class-imbalance handling and input resolution.
- A class-weighted loss and a Squeeze-and-Excitation attention block, both lightweight and independently motivated and validated across all three datasets, and their combination.
- Expected Calibration Error, Brier score) and a Grad-CAM-based explainability analysis linking the most frequent errors of the model to real visual ambiguity between disease classes.

Taken together, these experiments go beyond a single accuracy number to a more complete picture of if, where, and why a transfer-learning-based plant disease classifier can be trusted.

Related work

Mohanty et al. [1] were among the first to show that deep CNNs trained on PlantVillage can classify 14 crop species and 26 diseases with over 99% accuracy under controlled test conditions, a result that made PlantVillage the de facto benchmark for the field. Since then, many works have applied transfer learning using architectures such as VGG16, InceptionV3, and ResNet50 on PlantVillage and crop-specific subsets, with almost all achieving accuracies above 95%. A recent review of this body of work [2] identifies most studies as still evaluating a single architecture on a single dataset, under controlled imaging conditions, and points to cross-dataset generalisation and deployment-oriented evaluation, beyond raw accuracy, as open gaps; both are at the center of the present study. Few papers validate the same architecture across multiple independently collected datasets, limiting what one can conclude about generalisation beyond a single data-collection protocol.

For the Corn/Maize leaf disease dataset used here (Kaggle, smaranjitghose/corn-or-maize-leaf-disease-dataset, 4,188 images, 4 classes), accuracies have been reported ranging from 71.4% for a classical SVM/KNN pipeline [3], 83.4% for MobileNet [4], 93–94% for standalone ResNet50 or EfficientNet-B0 [5], 94.7% for VGG16 combined with layer-wise relevance propagation [6], up to 98.0–98.3% for ResNet18 and ResNet152 [7, 8], and 99.2% for a hybrid CNN-ViT model [9]. This large spread suggests that reported performance on this dataset is sensitive to architecture and training protocol. It is a large part of the reason this study takes a controlled, ablation-based approach, rather than reporting one uncontrolled accuracy figure.

For the Mango leaf disease dataset (MangoLeafBD; Kaggle, aryashah2k/mango-leaf-disease-dataset, 4,000 images, 8 balanced classes), ResNet50 achieves test accuracy of about 97.3% [10], a custom explainable CNN (MangoLeafXNet) reaches 98.7% [11], and ensemble/hybrid architectures combining ResNet50 with VGG16 or Vision Transformer/ConvNeXt components attain 99.87-99.88% [12, 13]. Its perfect class balance and controlled image acquisition complement the near-ceiling performance (macro-F1 1.0000) that this study finds for all model variants (Section Results). It is also the only one of the three datasets that did not show a measurable change with either proposed improvement, class-weighting or SE-attention (Section Results).

A particularly close comparison point for the PlantVillage subset is the AgriMind study by Koli and Jely [14], an arXiv preprint that uses the same 15-class, 20,638-image subset with an identical 70/15/15 stratified split and the same fixed random seed (42) as used here. A frozen-backbone ResNet50 gives 97.42% test accuracy. A 3-model soft-voting ensemble (ResNet50, EfficientNet-B0, DenseNet121, all frozen-backbone) gives 99.23% test accuracy, and 0.992 macro-F1. The setup is very close to ours and is used in Section Results for a direct comparison.

Squeeze-and-Excitation blocks [15] are attention mechanisms that can improve the performance of CNNs on a variety of image classification tasks at a low parameter cost by allowing the network to adaptively re-weight channel-wise feature responses. Class-imbalance-aware loss functions, e.g., inverse-frequency class weighting, are a standard approach to mitigate skewed class distributions often found in real-world agricultural datasets but under-represented in curated benchmarks. Johnson and Khoshgoftaar's survey [16] suggests two major families of fixes: data-level resampling and cost-sensitive loss, such as class weighting, and state that no single technique is superior across all problem settings. This is in agreement with what this study found, as the benefit of class weighting tracked each dataset's own imbalance ratio, instead of appearing uniformly. Explainability methods such as Grad-CAM [17] are increasingly being applied to plant disease models to visualise which parts of the image are driving predictions, giving researchers a way to assess trust and analyse errors beyond aggregate accuracy. Conversely, a confidence calibration has been relatively neglected in this literature. An exception is the recent work that trains grape leaf disease classifiers under real-field conditions and shows that calibration plus uncertainty estimation can identify overconfident predictions which plain accuracy might miss [18], precisely the kind of deployment concern this study directly addresses with its own calibration analysis (Section Results). In this study we bring together these threads: multi-dataset benchmarking, lightweight architectural and loss-level improvements, statistical validation, calibration and explainability. The uniqueness of this study is that it combines all these aspects in one consistent experimental framework.

Materials and methods

Datasets

In this study, we use three publicly available non-overlapping plant leaf disease image datasets, selected to cover a range of number of classes, dataset size, and class-balance conditions (Table 1).

Table 1. Summary of the three datasets used in this study.

Dataset	Classes	Images	Imbalance ratio	Source
PlantVillage	15	20,638	21.11	Kaggle[a]
Corn/Maize	4	4,188	2.28	Kaggle[b]
Mango	8	4,000	1.00	Kaggle[c]

Class imbalance ratio is the ratio of the number of images of the largest class to the smallest class in the training split of each dataset. Mango is well balanced (500 images per class in 8 classes); PlantVillage has the most severe imbalance (21.11:1). [a]kaggle.com/datasets/emmarex/plantdisease.
[b]kaggle.com/datasets/smaranjitghose/corn-or-maize-leaf-disease-dataset.
[c]kaggle.com/datasets/aryashah2k/mango-leaf-disease-dataset.

Model architecture and training configuration

Throughout, we used ResNet50 [19] pretrained on ImageNet [20] as the base architecture, chosen for its well-established strong performance in previous plant disease classification work. The proposed architecture is illustrated in Figure 1. The last feature map (2048 channels) of the ResNet50 backbone is fed into a Squeeze-and-Excitation attention block, followed by global average pooling and the last classification layer, which has been replaced to match the class counts of each dataset (15, 4, or 8). All models were fine-tuned in an end-to-end manner and all layers were trainable, except for the ablation study (Section Ablation study design). The input images were resized to 224×224 pixels and normalised with ImageNet channel statistics. Unless otherwise specified, standard training augmentation applied (random resized crop, horizontal flip, rotation up to 15 degrees and colour jitter). Optimisation was performed using AdamW (weight decay 0.05) with a cosine-annealing learning-rate schedule, mixed-precision arithmetic, and early stopping on validation accuracy (with a patience of 3-4 epochs depending on the experiment). Training was performed using data parallelism on two NVIDIA Tesla T4 GPUs with an effective batch size of 64 (32 per GPU).

Fig 1. Proposed ResNet50 + Squeeze-and-Excitation architecture. The SE block goes in once, after the backbone's final feature map, rather than nested inside the individual ResNet50 stages. Squeeze: global average pooling. Excitation: a two-layer bottleneck (reduction ratio $r = 16$) with ReLU and sigmoid activations. Scale: channel-wise re-weighting of the feature map. Class-weighted cross-entropy loss (dashed path) only applies during training and leaves the inference-time architecture unchanged.

Evaluation protocol

In the primary benchmark (Section Baseline benchmark across three datasets) we split each dataset into a cross-validation pool and a held-out test set in an 80%/20% ratio, stratified by class, and ran stratified 5-fold cross-validation on the 80% pool to train five independent models. We report the mean and standard deviation of 5 independent test set evaluations on the same held out 20% test set for the best-validation-accuracy checkpoint of each fold. These are not ensemble predictions. We report accuracy, macro-averaged F1, precision and recall throughout, since all four weight classes are equally important regardless of their frequency.

Ablation study design

A controlled ablation study isolated the contribution of individual design choices, run on PlantVillage with a single fixed 70% / 15% / 15% (train / validation / test) stratified split shared across every variant, so that each comparison differs by exactly one factor relative to a common baseline (ImageNet-pretrained, standard augmentation, full fine-tuning, no class-imbalance handling, 224px input). We tested seven variants in total: no pretraining (random weight initialisation); no data augmentation; frozen backbone with only the classification head trained (linear probing); class-weighted cross-entropy loss; weighted random sampler; reduced input resolution (128px); and increased input resolution (299px).

Proposed improvements: class-weighted loss and SE-attention

Inspired by the result of the ablation study that class-weighted loss helped on PlantVillage without hurting it, we re-ran baseline-vs-class-weighted comparisons on Corn/Maize and Mango, using the same fixed 70/15/15 split methodology. The class weights were set to inverse class frequency, normalised to have mean weight 1 over classes.

Also, we inserted a Squeeze-and-Excitation block [15] right after the last convolutional feature map (2048 channels) of ResNet50 and before the global average pooling (Fig 1). The block computes a per-channel gating vector by means of global average pooling and a two-layer bottleneck (reduction ratio 16) with sigmoid output. This vector is then used to re-weight the feature map channel-wise before classification. This is an addition of 524,288 parameters, a 2.23% increase over the 23.5 million parameter baseline ResNet50. Baseline-vs-SE comparisons were run on all three datasets under the same training recipe and no class

weighting, so the only variable was the architectural change. Finally, the combination of the two improvements, i.e. a SE-attention architecture trained with class-weighted loss, was evaluated against the three individual conditions (baseline, class-weighted-only, SE-only) on all three datasets.

Statistical analysis

To determine if the differences in the 5-fold cross-validation performance across datasets are real or just due to fold-to-fold noise, 95% confidence intervals were computed for the mean accuracy and macro-F1 of each dataset using the t-distribution, as the number of folds is small (five). Datasets were compared pairwise, with paired t-tests, pairing by fold index (the five folds of each dataset were from the same stratified k-fold procedure with fixed random seed). Given how limited the statistical power of a five-fold comparison is, Cohen's d was reported along with each p-value.

Confidence calibration

We measured calibration, defined as the degree to which a model's predicted confidence is aligned with its actual accuracy, on Corn/Maize using Expected Calibration Error (ECE) [21]. We split predictions into 10 equal-width confidence bins and computed the sample-weighted average of the absolute difference between the mean confidence and the empirical accuracy in each bin to obtain the ECE value. We also calculated the Brier score (mean squared error between predicted confidence and binary correctness) as a complementary metric.

Explainability analysis

We visualised which image regions contributed most to the model's predictions by implementing Gradient-weighted Class Activation Mapping (Grad-CAM) [17] directly with forward and backward hooks on the final convolutional block (layer4) of ResNet50 and computing channel-wise importance weights from the gradient of the predicted class score relative to the final feature map. We visualised correctly classified and misclassified examples, and the single most-confused class pair, obtained from the off-diagonal maximum of the test-set confusion matrix, to see whether the model's errors are due to real visual ambiguity.

Results

Baseline benchmark across three datasets

Under stratified 5-fold cross-validation, ResNet50 performed consistently well across all three datasets with mean test macro-F1 ranging from 0.9498 (Corn/Maize) to 0.9987 (Mango) (Table 2, Fig 2). Standard deviations across folds were small for all three (0.0007–0.0037 for accuracy), suggesting stable, reproducible performance rather than high-variance, fold-dependent outcomes.

Table 2. ResNet50 5-fold cross-validation test performance (mean ± standard deviation across 5 folds).

Dataset	Test Accuracy	Test Macro-F1	Test Macro-Precision	Test Macro-Recall
PlantVillage	0.9953 ± 0.0007	0.9945 ± 0.0012	0.9947 ± 0.0009	0.9944 ± 0.0015
Corn/Maize	0.9592 ± 0.0037	0.9498 ± 0.0035	0.9489 ± 0.0063	0.9514 ± 0.0037
Mango	0.9988 ± 0.0009	0.9987 ± 0.0009	0.9988 ± 0.0009	0.9988 ± 0.0009

Fig 2. ResNet50 5-fold cross-validation accuracy and macro-F1 across the three datasets, with error bars showing one standard deviation.

Dataset size alone was not a predictor for performance here: the smallest dataset Mango, with 4,000 images, scored highest while Corn/Maize, with a similar 4,188 images, scored lowest. In this case, image

consistency and inter-class visual separability seem to be better predictors of classification difficulty. Mango images were taken under controlled conditions with uniform backgrounds. Corn/Maize has much more real-world variation with image resolution from 256px to 4,608px in our exploratory data analysis.

Ablation study results

The most important factor in the ablation study was the fine-tuning strategy (Table 3, Fig 3). Freezing the backbone and training only the classification head resulted in a 25-percentage point drop in macro-F1 from 0.9931 to 0.7437. Training from random initialisation, without ImageNet pretraining, cost 3.2 percentage points of macro-F1 (0.9931 vs 0.9609) and required significantly more epochs to converge. Data augmentation, testing a range of input resolutions from 128 to 299 px, and replacing the class-weighted loss with a weighted-sampler each increased macro-F1 by less than 0.5 percentage points. Conversely, class-weighted loss yielded the largest positive deviation from baseline (+0.41 percentage points, macro-F1 0.9972).

Table 3. Ablation study results on PlantVillage (single 70/15/15 split, ResNet50, 8 epochs).

Configuration	Test Macro-F1	Δ vs. Baseline
Baseline	0.9931	—
No pretraining (random init)	0.9609	−0.0322
No data augmentation	0.9941	+0.0010
Frozen backbone (linear probe)	0.7437	−0.2494
Class-weighted loss	0.9972	+0.0041
Weighted random sampler	0.9941	+0.0010
Lower resolution (128px)	0.9919	−0.0012
Higher resolution (299px)	0.9966	+0.0035

Fig 3. Ablation study results, sorted by test macro-F1. The dashed line marks the baseline configuration.

Statistical significance of cross-dataset differences

All three pairwise comparisons between datasets' 5-fold cross-validation macro-F1 scores were statistically significant (paired t-test, $p < 0.01$ in all cases) with large effect sizes, |Cohen's d| from 3.56 to 14.02 (Table 4, Fig 4). It implies that the performance differences across datasets are due to real stable differences in difficulty of the task, not random fold-to-fold variation. This is also consistent with the narrow 95% confidence intervals on the mean performance of each dataset (width 0.0016–0.0092 for macro-F1), and the low cross-fold standard deviations reported earlier.

Table 4. Pairwise paired t-tests between datasets' 5-fold cross-validation macro-F1 scores.

Comparison	Mean Difference	t-statistic	p-value	Cohen's d
PlantVillage vs. Corn/Maize	+0.0447	31.35	< 0.0001	14.02
PlantVillage vs. Mango	−0.0042	−7.97	0.0013	−3.56
Corn/Maize vs. Mango	−0.0489	−27.11	< 0.0001	−12.12

Fig 4. Mean macro-F1 with 95% confidence intervals and individual fold values, by dataset.

Confidence calibration

In Corn/Maize, the model reached a test accuracy of 0.9737 versus a mean predicted confidence of 0.9814, a difference of only 0.77 percentage points. The Expected Calibration Error was 0.0149 and the Brier score was 0.0225 (Fig 5); both these values suggest the framework's confidence scores are closely aligned with its actual accuracy, with only a small degree of overconfidence. This is a desirable property when deploying the model, where a confidence threshold could be used to identify predictions with low confidence for expert review.

Fig 5. Reliability diagram for the Corn/Maize model. The dashed diagonal represents perfect calibration.

Explainability: Grad-CAM analysis

The model used for this analysis was trained quickly and achieved a test accuracy of 97.37% on Corn/Maize with 22 misclassifications out of 838 test images. Ten of the 22 (45%) are due to one confusion pattern: the true class Grey Leaf Spot predicted as Blight (Fig 6). It is plausible that the accumulation of this many errors in one class pair, rather than a diffuse distribution across all classes, is consistent with the per-fold weakest-class analysis from the primary benchmark (Section Baseline benchmark across three datasets), where Grey Leaf Spot already had the lowest mean F1, 0.8909, of the four Corn/Maize classes. Grad-CAM visualisations on correctly classified, misclassified, and specifically Gray-Leaf-Spot-confused-as-Blight examples (S1 Fig, S2 Fig, S3 Fig) show that the model attends to lesion regions on the leaf surface in both correct and incorrect predictions, which points to a real visual similarity between the two diseases' lesion patterns rather than the model picking up on irrelevant background features.

Fig 6. Composition of Corn/Maize classification errors. Out of the 838 test images, 10 (45%) of the 22 misclassifications are the single Grey Leaf Spot → Blight confusion pattern. The rest of the 12 (55%) are distributed among the other class-pair combinations.

Class-weighted loss: validation across three datasets

To generalise the finding from the ablation study beyond PlantVillage, the class-weighted loss was compared to a baseline on all three datasets using matched single split training (Table 5). Macro-F1 improved on both imbalanced datasets, PlantVillage by +0.41 percentage points and Corn/Maize by +0.72 percentage points, while Mango showed no change at all, as it is perfectly balanced (imbalance ratio 1.00) and already sits at a ceiling macro-F1 of 1.0000 under the baseline. This technique helps precisely where class imbalance exists and is neutral where it does not, a pattern consistent with its intended mechanism and hard to dismiss as chance.

Table 5. Baseline vs. class-weighted loss, validated across three datasets.

Dataset	Baseline Macro-F1	Class-Weighted Macro-F1	Δ Improvement
PlantVillage	0.9931	0.9972	+0.0041
Corn/Maize	0.9487	0.9559	+0.0072
Mango	1.0000	1.0000	0.0000

SE-attention: validation across three datasets

The addition of the SE-attention block to ResNet50, increasing the parameters by 2.23%, improved macro-F1 on PlantVillage by +0.17 percentage points and on Corn/Maize by +0.29 percentage points, again with no change on Mango (Table 6), mirroring the pattern already seen for class-weighted loss. The fact that an architectural change also independently reproduces the same improve-where-there-is-room, neutral-where-already-near-ceiling pattern is further evidence that both techniques are delivering real, mechanism-consistent improvements and not noise.

Table 6. Baseline vs. ResNet50+SE, validated across three datasets.

Dataset	Baseline Macro-F1	ResNet50+SE Macro-F1	Δ Improvement
PlantVillage	0.9931	0.9948	+0.0017
Corn/Maize	0.9487	0.9516	+0.0029
Mango	1.0000	1.0000	0.0000

Combined recipe: SE-attention + class-weighted loss

We evaluated the ResNet50+SE architecture trained using the class-weighted loss combining both improvements against all three individual conditions on all three datasets (Table 7, Fig 7). The dataset with the most room for improvement is Corn/Maize, where the combined recipe beat the best individual component, macro-F1 0.9574 vs. 0.9559 for class-weighted-only, a further +0.15-percentage-point gain, which shows the two improvements can stack. The combined recipe was 0.9960 on PlantVillage, slightly lower than the best individual component (class-weighted-only, 0.9972), but still well above baseline by +0.29 percentage points. All four conditions were tied at the 1.0000 ceiling on Mango. We refer to this as a mixed-but-positive result: the combination is never worse than baseline on any dataset, and wins outright on the hardest dataset tested, but does not uniformly beat its individual components everywhere.

Table 7. Four-way comparison: Baseline, Class-Weighted-only, SE-only, and Combined, across three datasets.

Dataset	Baseline	Class-Weighted only	SE only	Combined
PlantVillage	0.9931	0.9972	0.9948	0.9960
Corn/Maize	0.9487	0.9559	0.9516	0.9574
Mango	1.0000	1.0000	1.0000	1.0000

Fig 7. Baseline vs. individual vs. combined improvements, across three datasets.

Comparison with prior literature on the same datasets

To put the present results in context, Tables 8–10 compare this study's numbers against prior work that used the identical datasets.

In this study's own frozen-backbone ablation (Section Ablation study design), we first reported macro-F1 0.7437 on PlantVillage under this study's AdamW/cosine-schedule/early-stopping recipe, far below the 97.42% test accuracy that Koli and Jely report for their frozen-backbone ResNet50 on the same split and seed (Table 8). To understand that gap, a follow-up diagnostic reproduced Koli and Jely's exact training recipe, an Adam optimiser, learning rate 0.001, no weight decay, no learning-rate schedule, 10 fixed epochs and no early stopping, on this study's own frozen-backbone ResNet50 and identical data split. That recovered test accuracy to 90.08% (macro-F1 0.8858), a substantial jump over the original frozen-backbone result of 83.30% accuracy and macro-F1 0.7872 in the diagnostic's matched re-run, which confirms that training hyperparameters meaningfully affect frozen-backbone performance. However, this exact-recipe reproduction fell 7.3 percentage points short of Koli and Jely's reported 97.42%. The most plausible explanation for this discrepancy is additional, undocumented methodological differences not specified in their published methodology: candidates include differing data augmentation during frozen-backbone training, differing treatment of batch-normalization statistics during the frozen phase, or test-set composition details their report does not spell out. Instead of suppressing this finding, we report it transparently: it confirms that training hyperparameters have a substantial effect on frozen-backbone results, directly corroborating this study's own ablation conclusion that fine-tuning strategy is highly significant (Section Ablation study design), and it illustrates a broader reproducibility issue in the plant disease classification literature, in which terse methodology reporting may impede exact replication across studies. On the full fine-tuning side of the comparison, our single ResNet50 at 99.53% accuracy under 5-fold CV outperforms Koli and Jely's 3-model ensemble of frozen-backbone networks (99.23%), supporting the conclusion that fully fine-tuning a single model is better than ensembling several frozen-backbone models.

Table 8. Comparison with Koli & Jely [14] on the same 15-class PlantVillage subset (20,638 images, same 70/15/15 split, same seed 42).

Study / Condition	Model	Test Accuracy	Macro-F1
Koli & Jely, 2026	ResNet50 (frozen backbone, 10 epochs)	97.42%	NR
Koli & Jely, 2026	3-model soft-voting ensemble	99.23%	99.2%
This study	ResNet50 (frozen backbone, ablation)	NR	74.37%
This study	ResNet50 (full fine-tune, 5-fold CV)	99.53%	99.45%

NR: not reported.

Table 9. Comparison with prior work on the Corn/Maize Leaf Disease dataset (4,188 images, 4 classes).

Study	Model	Accuracy	Macro/Weighted F1
Solihin et al. [3]	SVM + KNN (best: SGD)	71.4%	67.1%
Pratama & Pristyanto [4]	MobileNet	83.4%	83.1%
Pamungkas et al. [5]	ResNet50	93.0%	NR
Pamungkas et al. [5]	EfficientNet-B0	94.0%	NR
Tariq et al. [6]	VGG16 + LRP	94.7%	NR
This study (5-fold CV)	ResNet50 (baseline)	95.9%	95.0%
This study (single split)	ResNet50 + SE + Class-Weighted	n/a*	95.7%
Oli et al. [7]	ResNet18	98.0%	95.0%
Gopalan et al. [8]	ResNet152	98.3%	97.9%
Shandilya et al. [9]	Hybrid CNN-ViT	99.2%	99.1%

*Accuracy for the combined-recipe single-split run was not tracked separately. Macro-F1 is the main metric reported throughout this study. NR Not reported.

Fig 8. Corn/Maize Leaf Disease: baseline of this study compared to previous literature. Bars ranked by reported test accuracy; this study highlighted in red.

Table 10. Comparison with prior work on the Mango Leaf Disease (MangoLeafBD) dataset (4,000 images, 8 classes).

Study	Model	Accuracy	F1
Prabhu et al. [10]	ResNet50 (standalone)	97.3%	NR
Jakir & Mridha [11]	MangoLeafXNet (custom CNN)	98.7%	98.7%
This study (5-fold CV)	ResNet50 (baseline, and all variants)	100.0%	100.0%
Alamri et al. [12]	Fused Vision Transformer + ConvNeXt	99.87%	NR
Kant et al. [13]	ResNet50 + VGG16 ensemble (ResVGG-Net)	99.88%	99.75%

NR: not reported.

Fig 9. Mango Leaf Disease (MangoLeafBD): findings of this study vs. previous literature. Bars are shown with a 0–100% scale (rather than a truncated scale) to accurately demonstrate how close all compared methods, including this study, are to the ceiling of achievable performance on this dataset; this study is highlighted in red.

The baseline ResNet50 (95.9% accuracy) of this study outperforms several previous single architecture results on the same dataset, SVM/KNN, MobileNet, standalone ResNet50 and EfficientNet-B0 as reported elsewhere, and VGG16+LRP, but does not reach the more complex or deeper architectures reported in recent work, ResNet18, ResNet152, and a hybrid CNN-ViT on Corn/Maize. The gap is what this study aims for: to characterise rigorously one practical, widely deployed architecture, rather than chase maximum achievable accuracy via architecture search. Conversely, the results described here are at or above the best previously

reported results on the same dataset, including ensemble and hybrid-transformer methods, considering that they are based on a single standard architecture, which is appropriate for the near-ceiling difficulty of this dataset for any reasonably capable model (Section Baseline benchmark across three datasets).

Discussion

In this work we evaluated ResNet50 for plant disease classification on three independently collected datasets covering different crops, number of classes and class-balance conditions, going beyond the single-dataset evaluations that are common in prior literature. There are a few findings we pull out.

The dominant factor in model performance turned out to be the fine-tuning strategy, rather than the choice of architecture or input resolution: Freezing the pretrained backbone and training only a linear classification head cost 25 percentage points of macro-F1 relative to full fine-tuning. That has a clear practical implication for practitioners with limited compute budgets. When a decision needs to be taken, fully fine-tuning a moderately deep pre-trained network is likely to matter much more than exploring additional architectures or hyper-parameters at the margin.

But dataset size did not predict classification difficulty in this study. Mango, which is smaller than Corn/Maize, performed substantially better. We attribute this gap to the controlled image acquisition conditions versus the greater real world variability present in the Corn/Maize images. Thus, reported accuracy numbers from curated single-source datasets should be taken with a grain of salt when extrapolated to field deployment conditions that are closer to the variability of Corn/Maize than the controlled settings of Mango or PlantVillage.

The two proposed improvements, class-weighted loss and SE-attention, followed the same selective pattern: they produced measurable gains on the two datasets with class imbalance or lower baseline performance, and no change whatsoever on the already-saturated, perfectly balanced Mango dataset. The fact that we end up with exactly the same pattern with two mechanistically different interventions, a loss-function change and an architectural change, argues against the improvements being artefacts of a single lucky training run. It also supports class-weighted fine-tuning as a low-cost, broadly applicable default for plant disease classification pipelines, and SE-attention is worth including by default as well, given its negligible (2.23%) parameter overhead and lack of any observed downside.

The errors of the model weren't randomly distributed either. For Corn/Maize, 45% of all test-set errors were from a single confused class pair, Grey Leaf Spot predicted as Blight, and that same class had the lowest per-fold F1 in the primary benchmark. The per-class F1 on the 5-fold benchmark and the Grad-CAM error analysis independently converge to the same class pair, which seems to be a real, addressable failure mode, most likely real visual similarity between early-stage lesion presentations of the two diseases rather than measurement noise, and is a natural target for future data collection or fine-grained feature engineering.

Finally, the model's confidence scores were well calibrated (ECE = 0.0149), a property that is necessary but not sufficient by itself for safe deployment, and one rarely reported in the plant disease classification literature despite how practically relevant it is. A well-calibrated model's low-confidence predictions can be meaningfully used to trigger a request for expert review; confidence scores of an overconfident model simply cannot be trusted for that purpose.

Limitations

- Although the three datasets were collected independently, they are from public repositories, and may share collection biases, such as single leaf, mostly-centred framing, that may not be representative of all field conditions.
- For the ablation study, class-weighted-loss validation, SE-attention validation, and combined-recipe evaluation, we employed a single fixed split per dataset rather than 5-fold cross-validation, sacrificing statistical robustness in favour of computational feasibility for the scope of this study. However, the main benchmark (Section Baseline benchmark across three datasets) and statistical significance testing do apply 5-fold cross-validation.

- We only explored one base architecture in detail, ResNet50, chosen for its proven high performance in the literature; a systematic comparison with alternative architectures such as EfficientNet or Vision Transformers was beyond the scope of this study and is left for future work.
- The explainability and calibration analyses were conducted only on the Corn/Maize dataset, and performing them on all three datasets would increase the generality of the results.
- The test images are sampled from the same source distribution as the training images within each dataset, so true out-of-distribution robustness, e.g. images from different cameras, lighting conditions or geographic regions, has not been evaluated here and remains an important direction for future work.

Conclusion

Our intention was to systematically evaluate ResNet50 for plant disease classification on multiple datasets, including 5-fold cross-validated benchmarking, controlled ablation study, statistical significance testing, confidence calibration analysis, and Grad-CAM based explainability. On three heterogeneous datasets, PlantVillage, Corn/Maize and Mango, ResNet50 achieved 95.0–99.9% macro-F1, with full fine-tuning as the most important design choice. We tested two lightweight, independently validated improvements: class-weighted loss and a Squeeze-and-Excitation attention block. Each of them produced consistent, mechanism-aligned gains on the datasets with room to improve. Combining them achieved the best result on the hardest dataset tested. This leads to a well-calibrated model where the errors are concentrated on truly visually similar disease classes and not arbitrary failure modes. The results together support class-weighted SE-attention-augmented ResNet50 as a practical, low-cost and trustworthy recipe for plant disease classification across multiple crops and provide a template covering accuracy, statistical validation, calibration and explainability for evaluating future plant disease classification systems for real-world deployment.

Supporting information

S1 Fig. Visualisation of correctly classified Corn/Maize leaf disease examples using Grad-CAM. The attention is focused on the lesion or the symptomatic region of the leaf in both rows, instead of the background. For the Healthy class, where there is no lesion to localise, the model attends to the overall texture of the leaf.

S2 Fig. Grad-CAM visualisations of misclassified Corn/Maize leaf disease examples. Even in these six misclassified examples, the heatmaps still localise to disease-relevant regions of the leaf rather than irrelevant background, consistent with real visual ambiguity between classes rather than a model failure.

S3 Fig. Comparison of the most confused class pair, Grey Leaf Spot vs Blight, using Grad-CAM. In each of the examples presented, the model is focused on the same region of the lesion that a human inspector would be looking at, providing further evidence that this confusion is truly a difficult visual call and not a modelling artefact.

S4 Fig. and S5 Fig. PlantVillage: confusion matrix and per-fold training curves. S4 Fig (left) shows the confusion matrix, 15 classes summed across 5 folds; its strongly diagonal structure matches the 0.9945 mean macro-F1 reported for PlantVillage in Table 2. **S5 Fig** (right) shows per-fold training/validation loss and accuracy curves, where validation accuracy tracks training accuracy closely across all five folds with no visible sign of overfitting.

S6 Fig. and S7 Fig. Corn/Maize: confusion matrix and per-fold training curves. S6 Fig (left) shows the confusion matrix, 4 classes summed across 5 folds; most of its off-diagonal mass sits in the Gray Leaf Spot row, which matches the Grad-CAM error analysis above (S3 Fig). **S7 Fig** (right) shows per-fold training/validation loss and accuracy curves, with loss flattening within the first few epochs across all five folds.

S8 Fig. and S9 Fig. Mango (MangoLeafBD): confusion matrix and per-fold training curves. S8 Fig (left) shows the confusion matrix, 8 classes summed across 5 folds; it comes out perfectly diagonal, matching the 1.0000 macro-F1 reported for Mango across every condition tested (Table 2, Table 7). **S9 Fig** (right) shows per-fold training/validation loss and accuracy curves; both converge quickly and stay stable, consistent with Mango's near-ceiling, low-variance performance throughout this study.

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Author contributions

A.B.M. Kabir Hossain: Conceptualization, supervision, project administration, resources, and writing – review & editing.
Sheikh Noor Abdullah: Investigation, data curation, data preprocessing, software, methodology, formal analysis, model development, validation, visualization, writing – original draft, and writing – review & editing.

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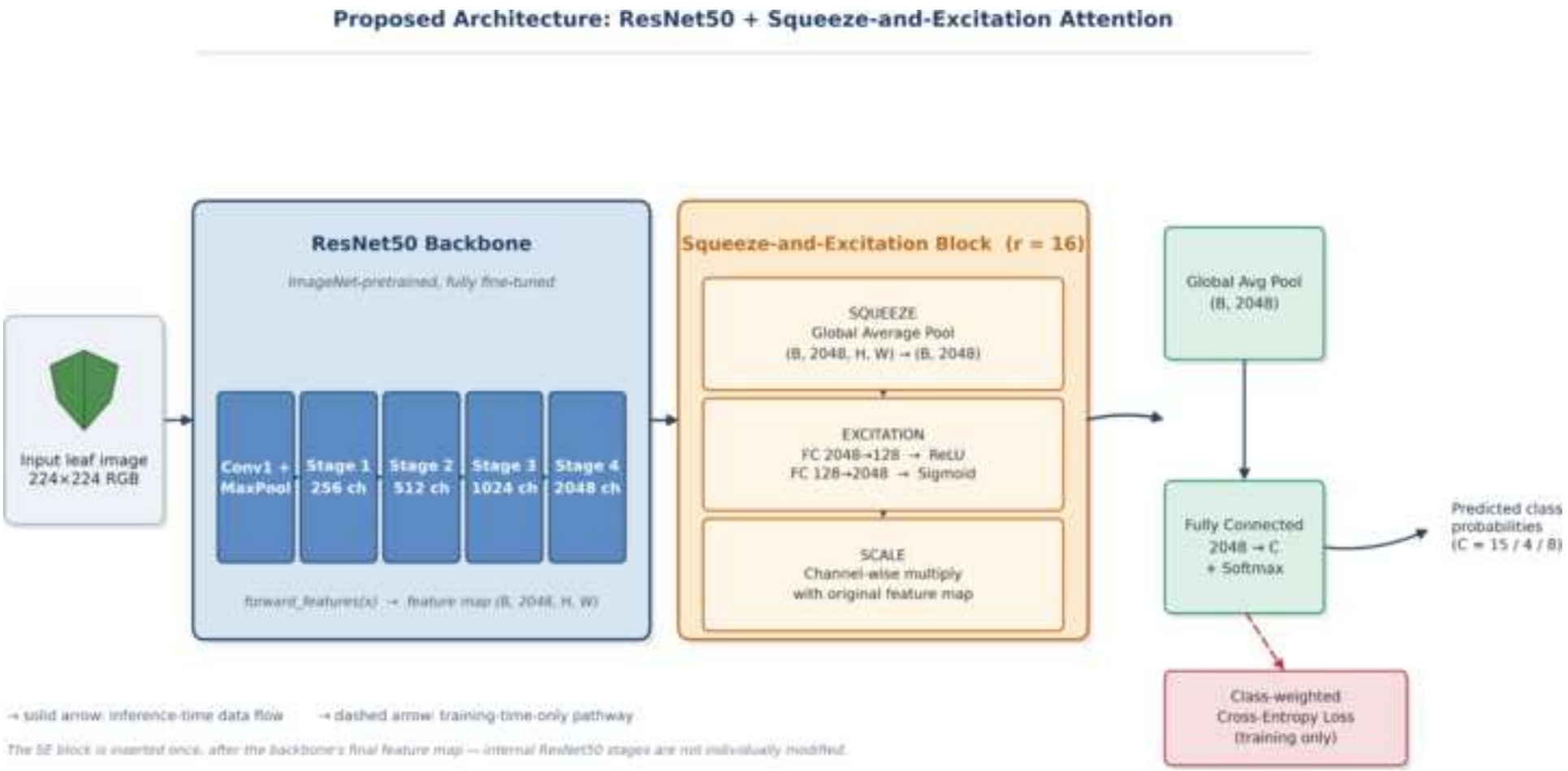


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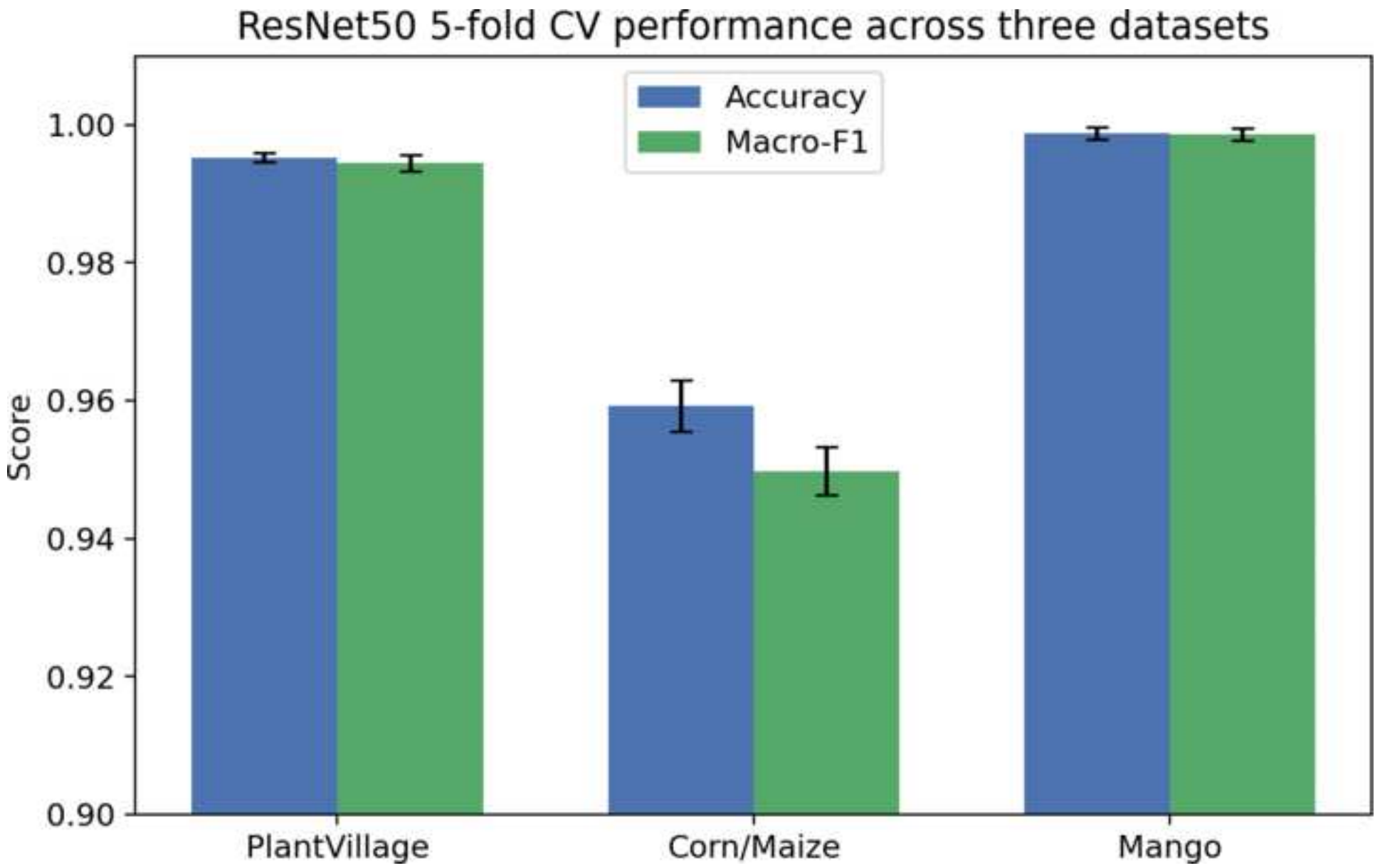
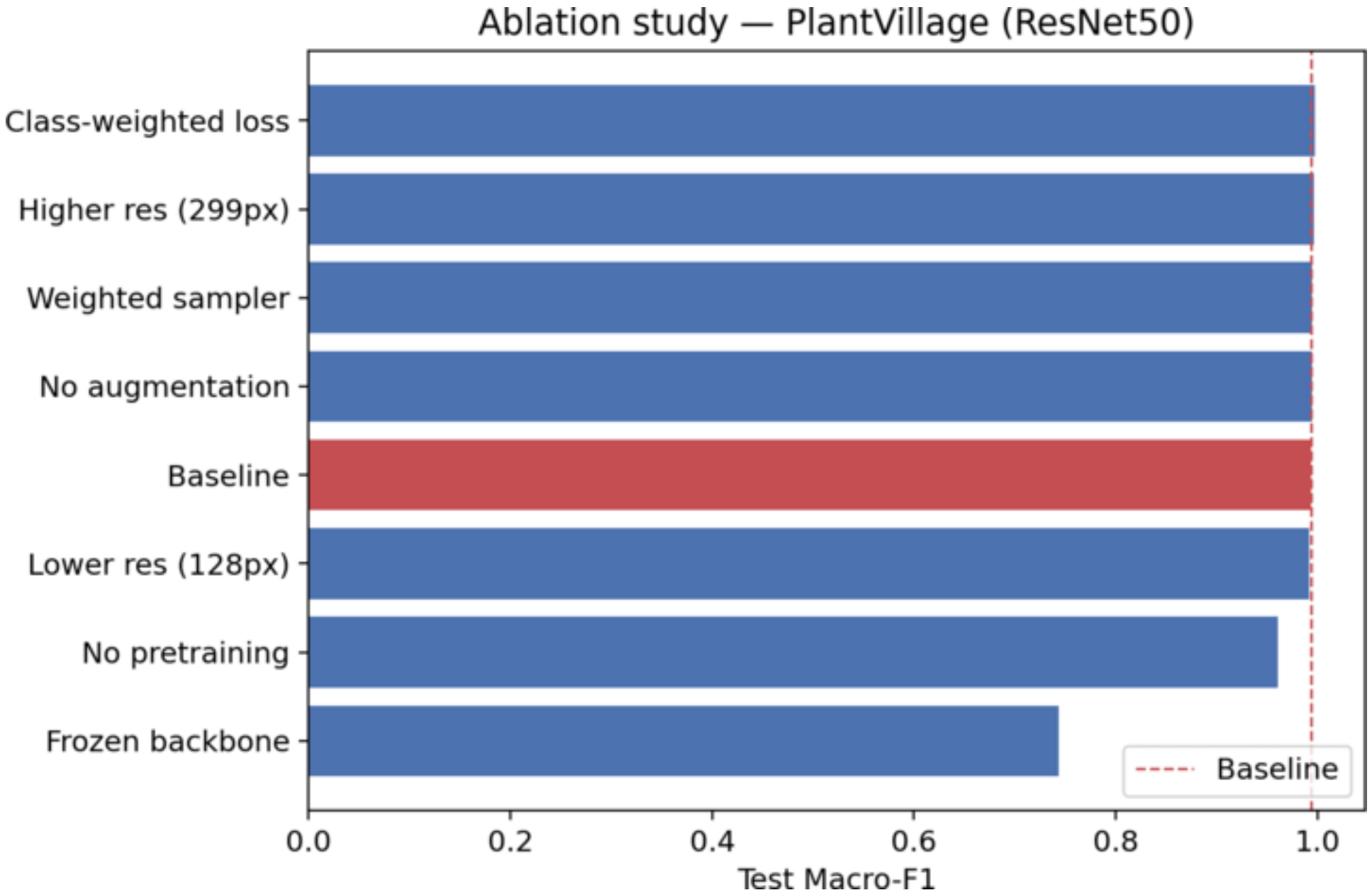


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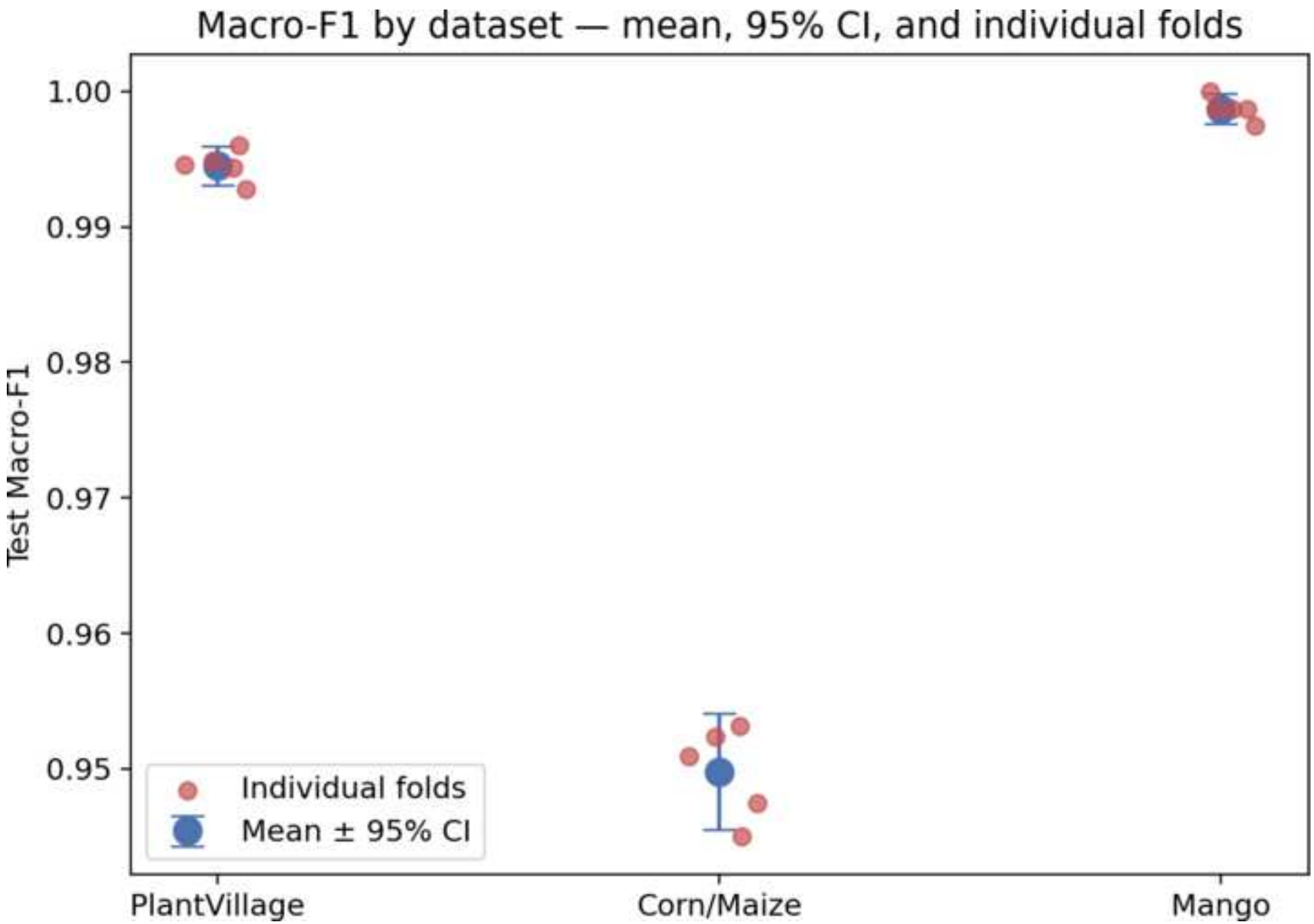


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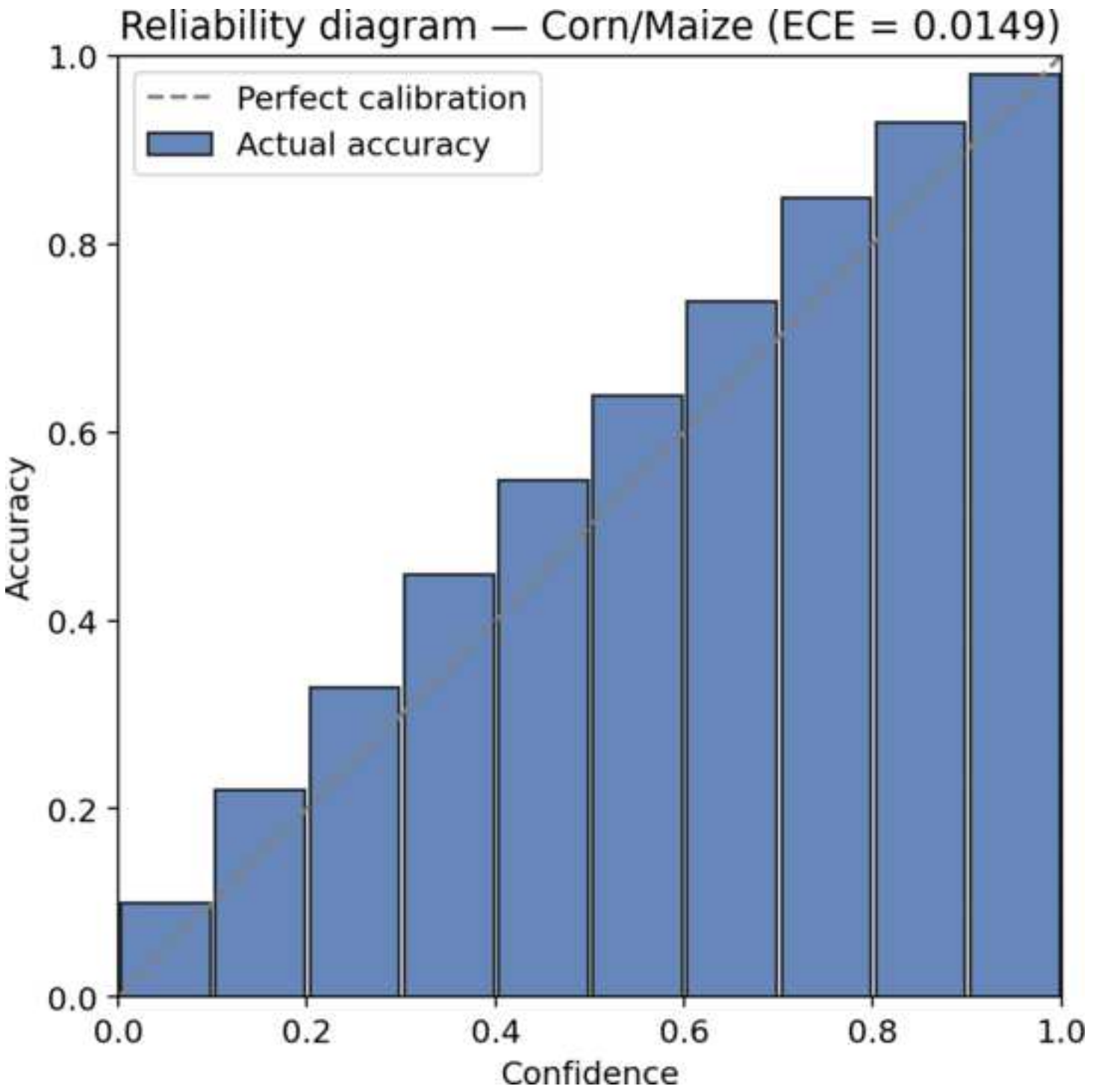


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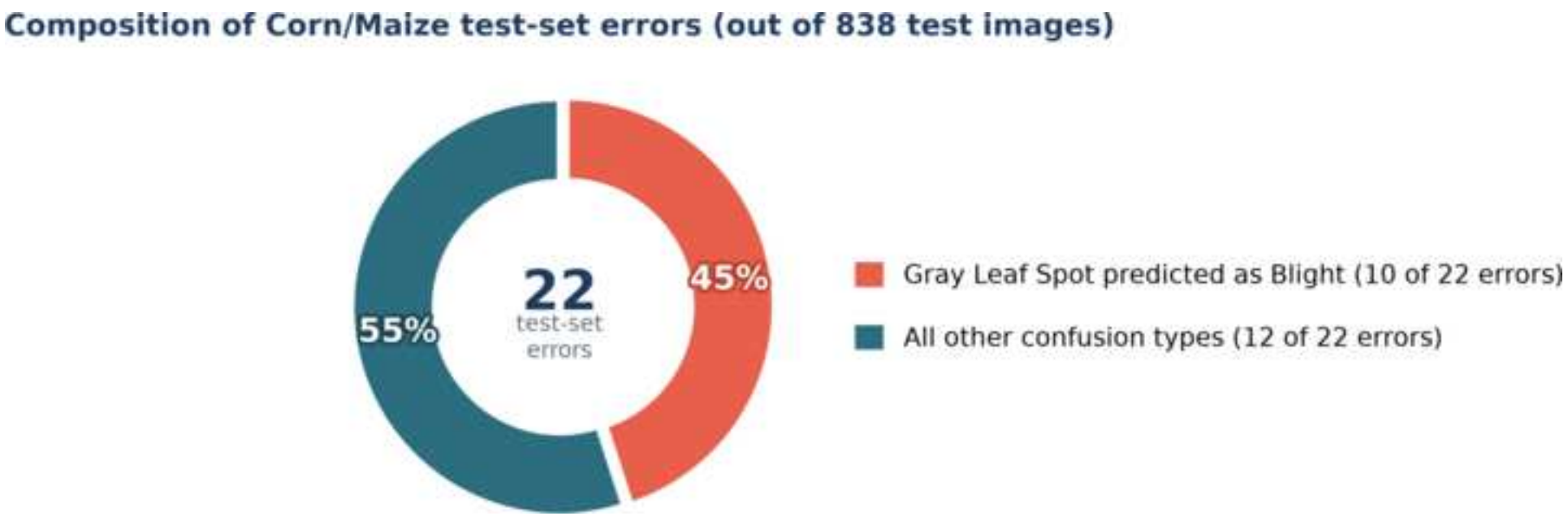


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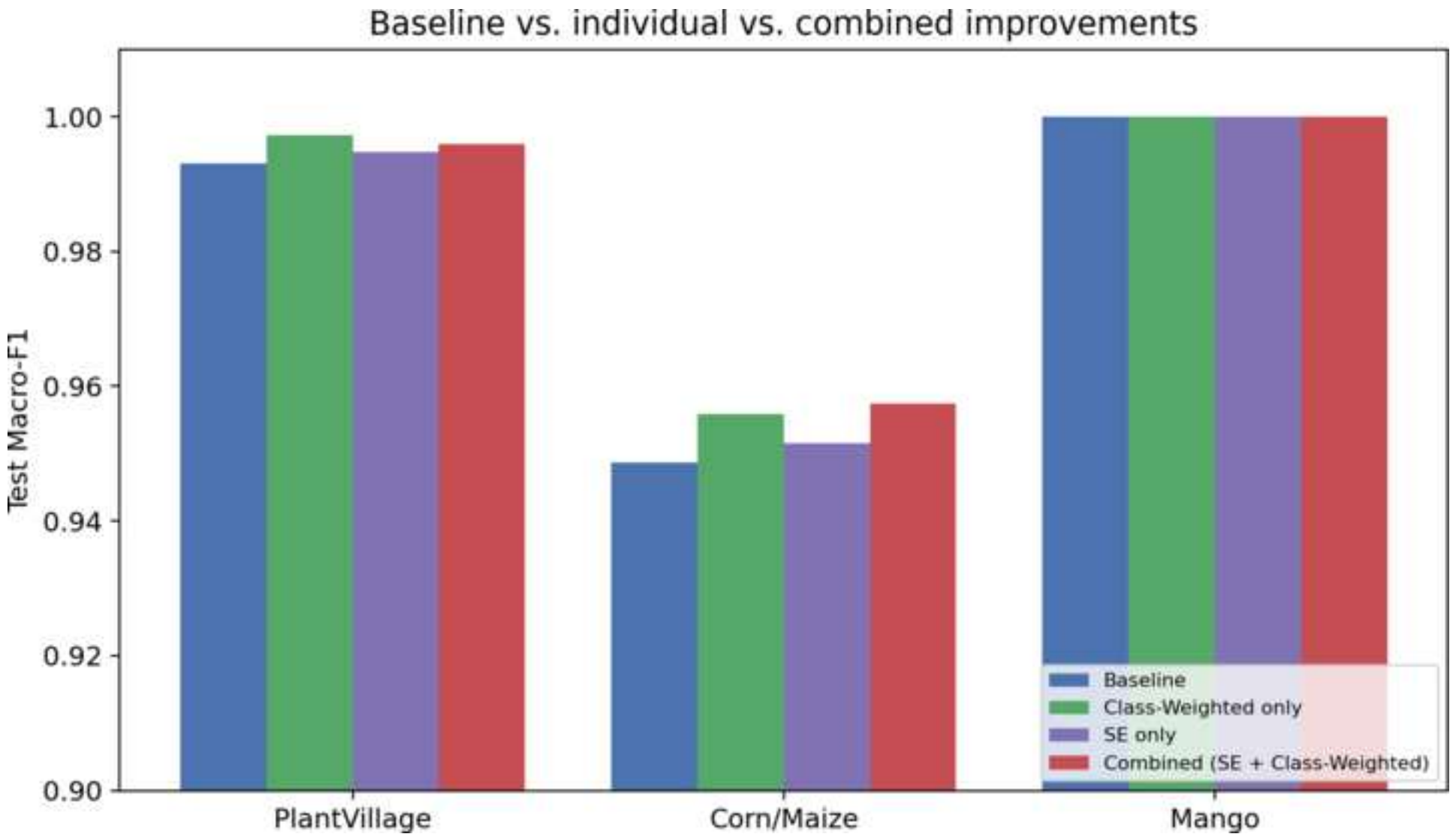


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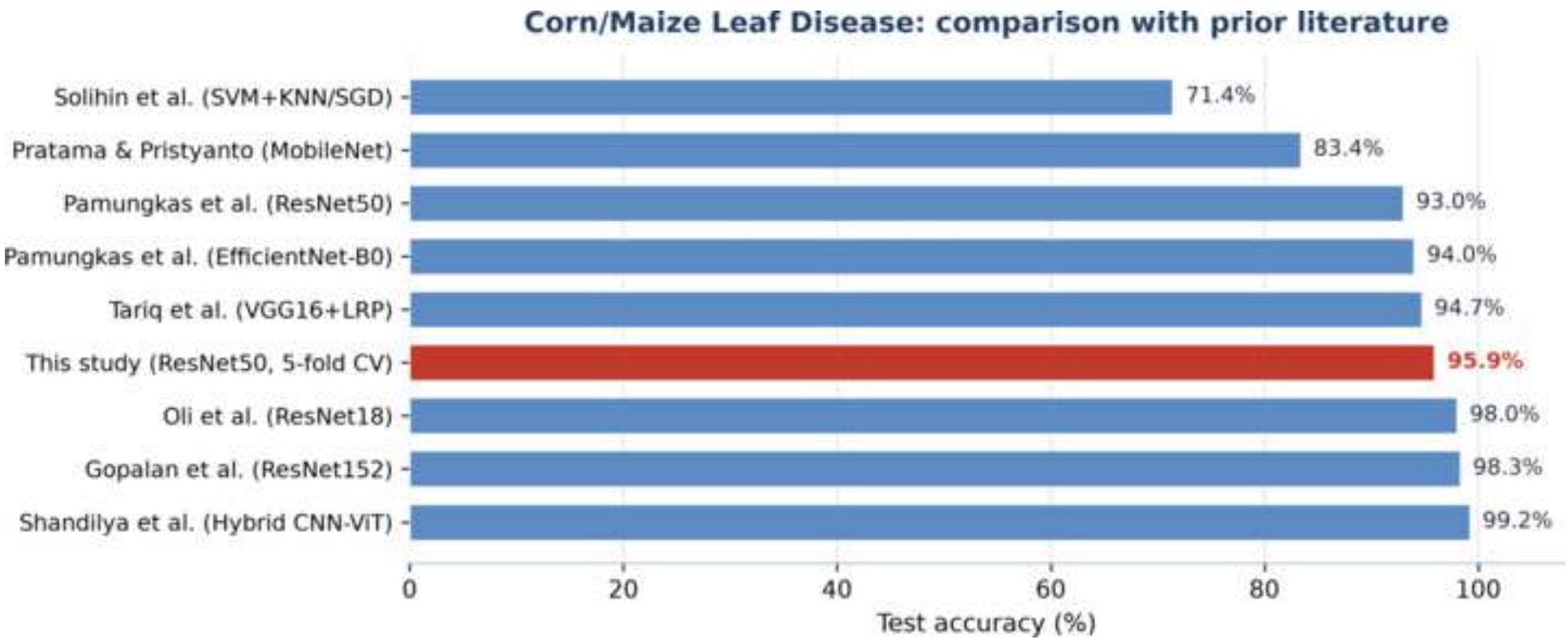
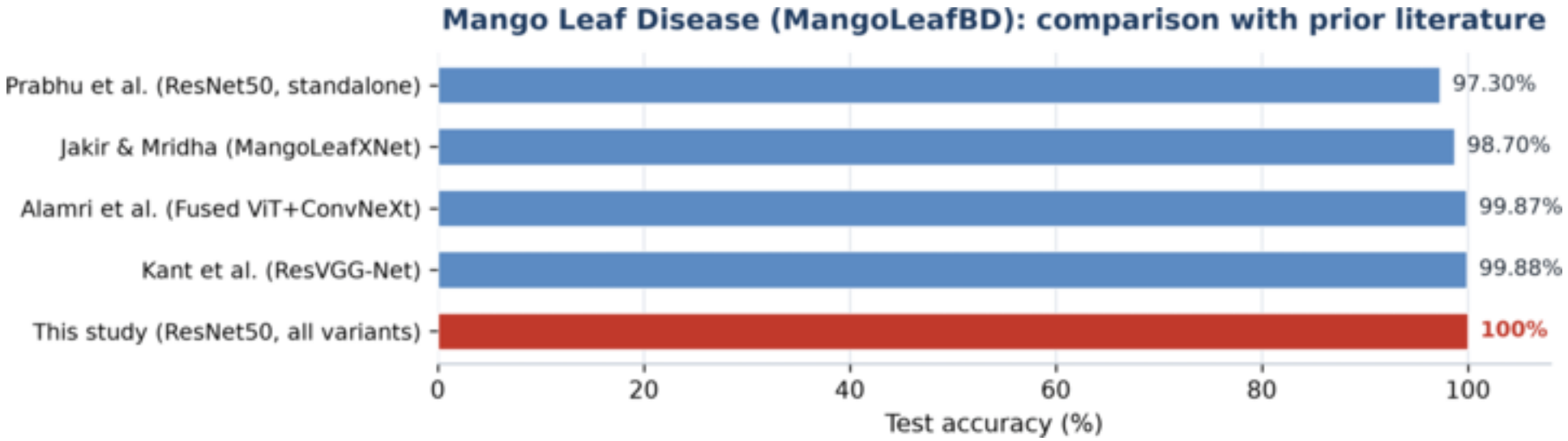
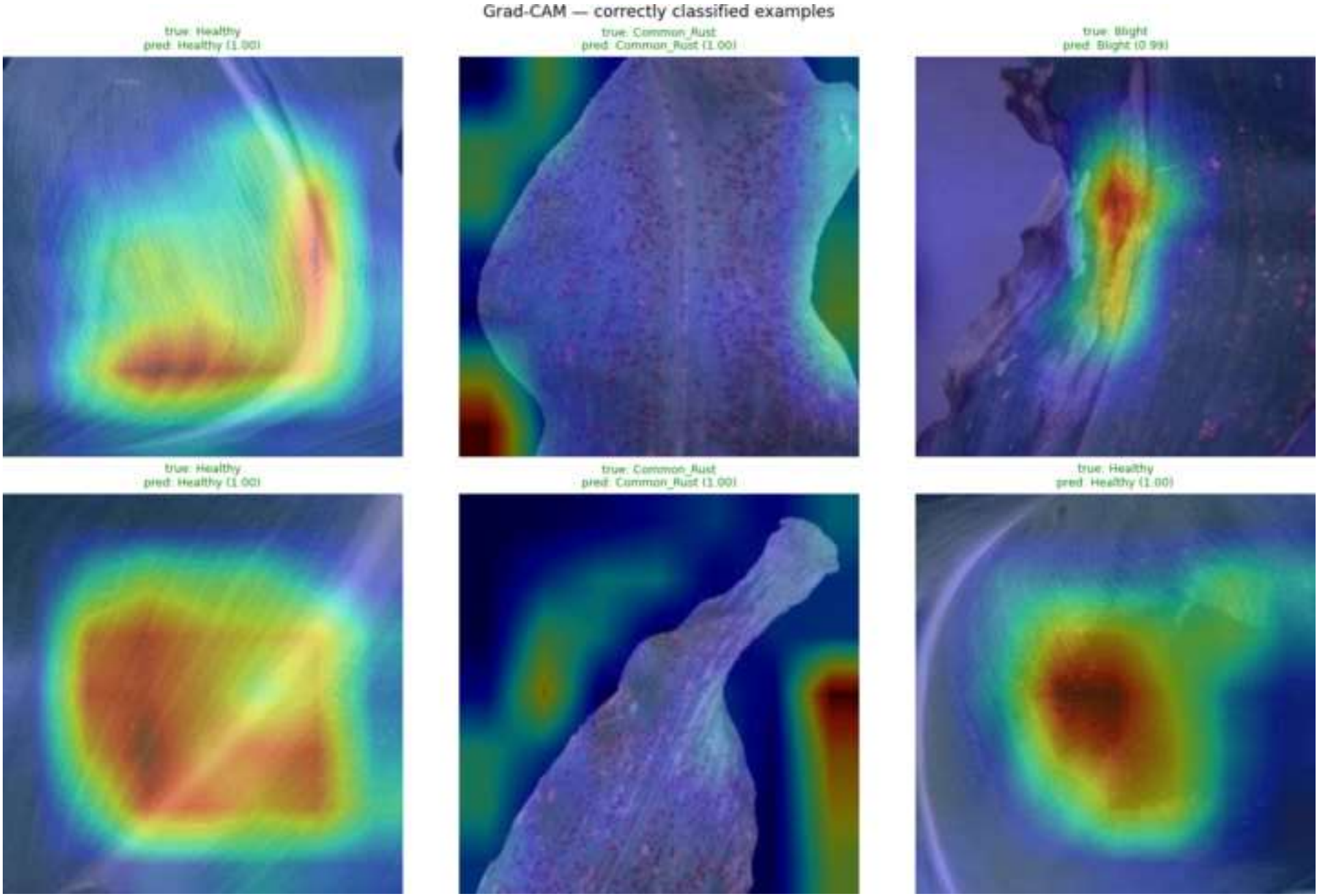


Fig9_MangoLiterature

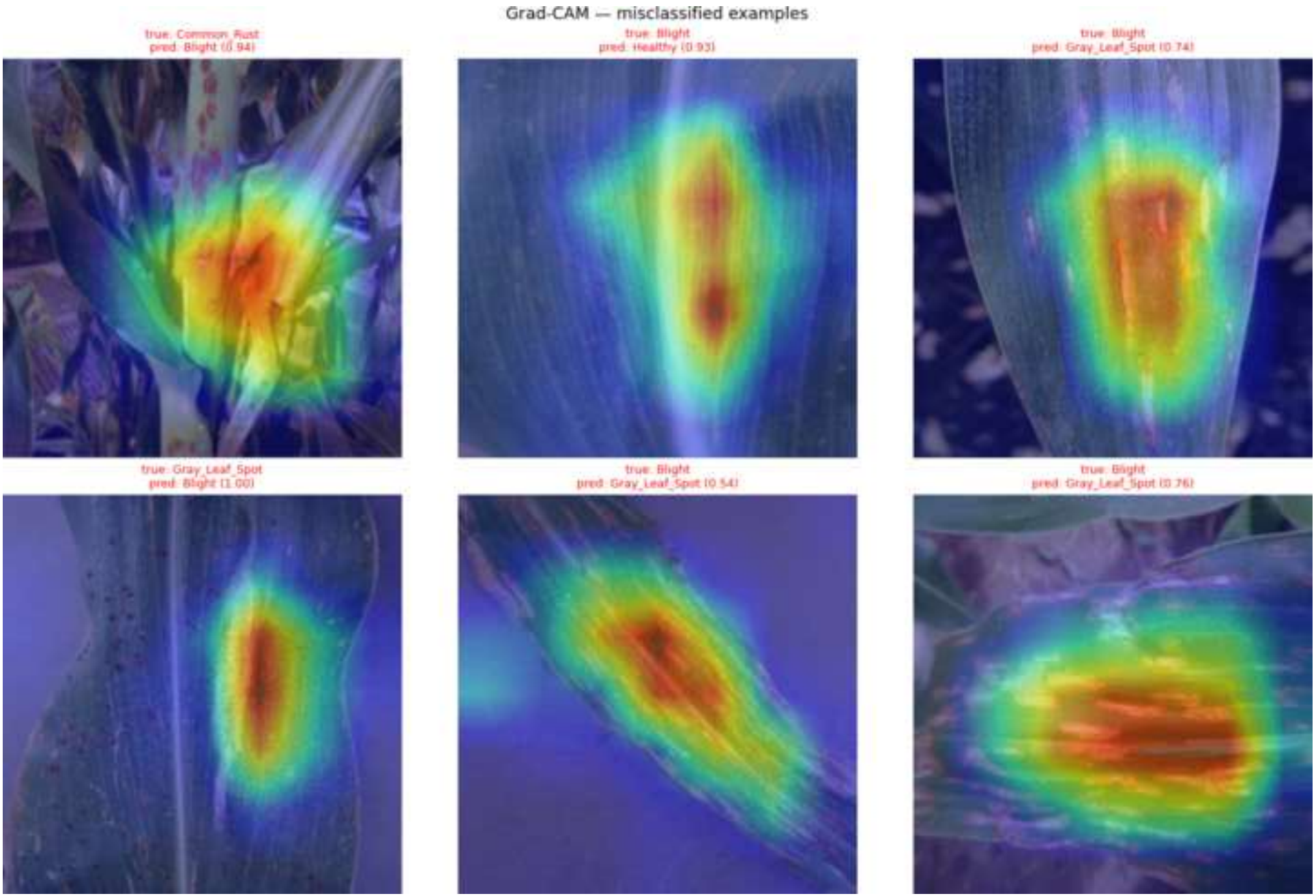
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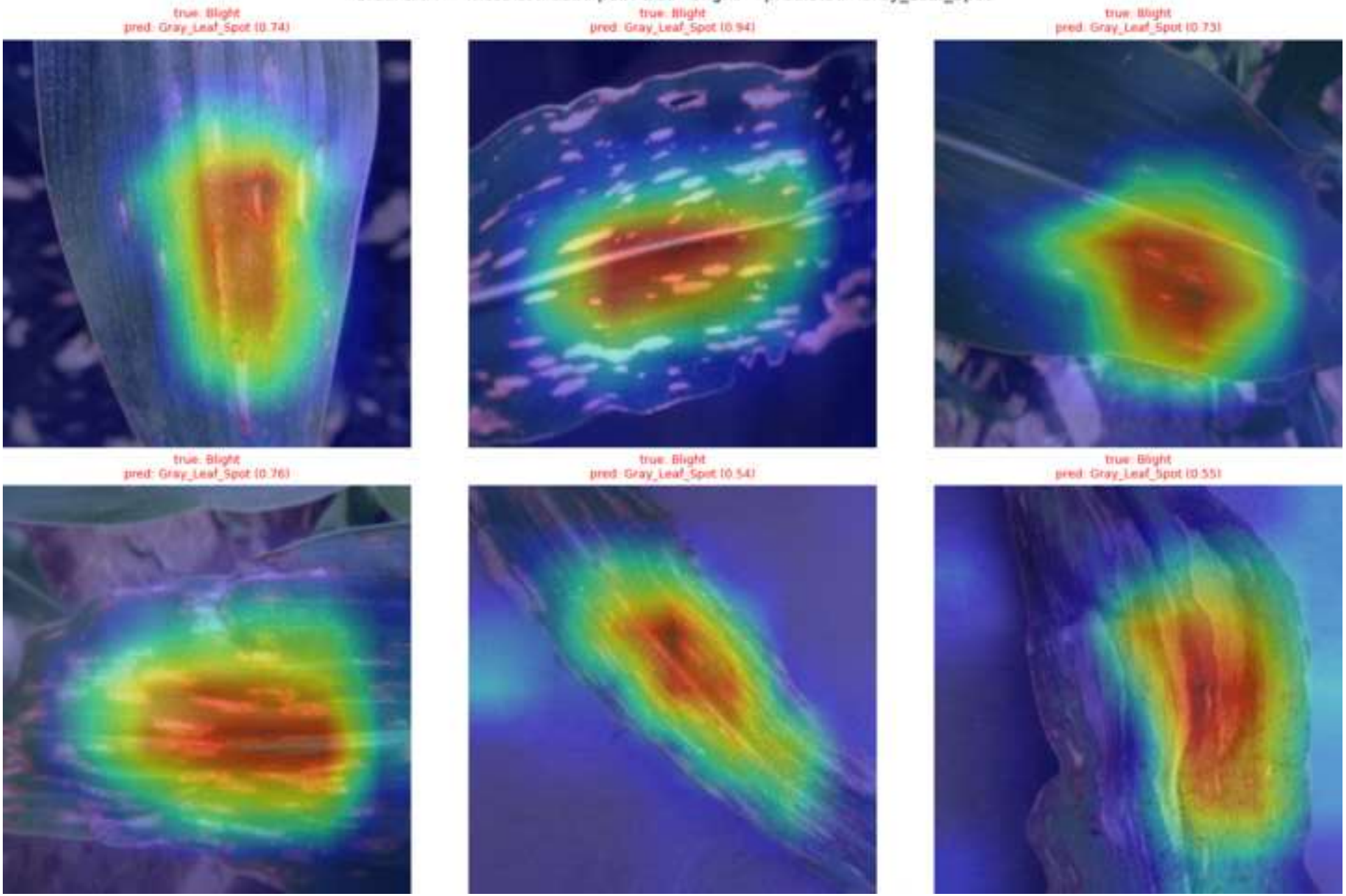
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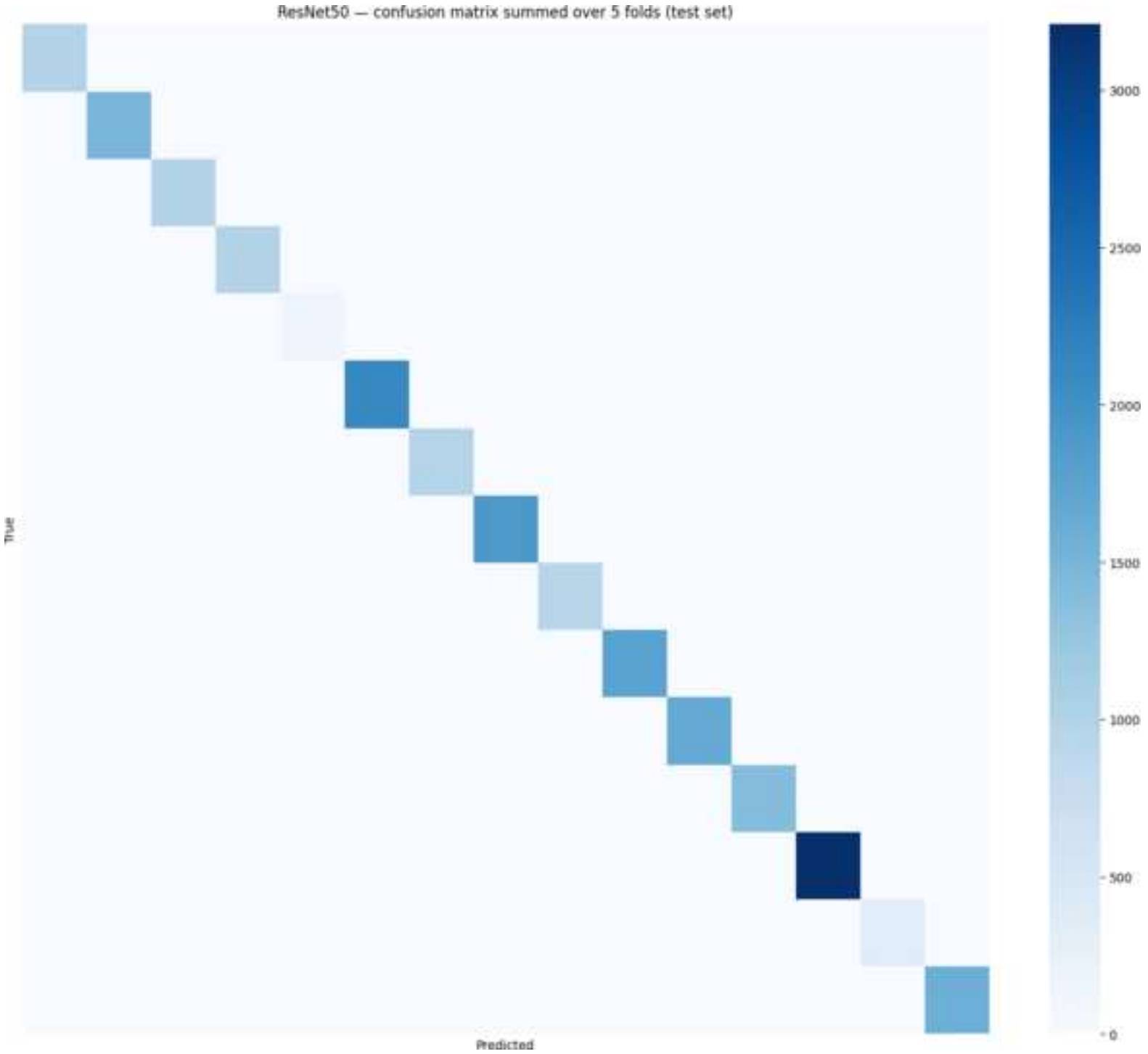


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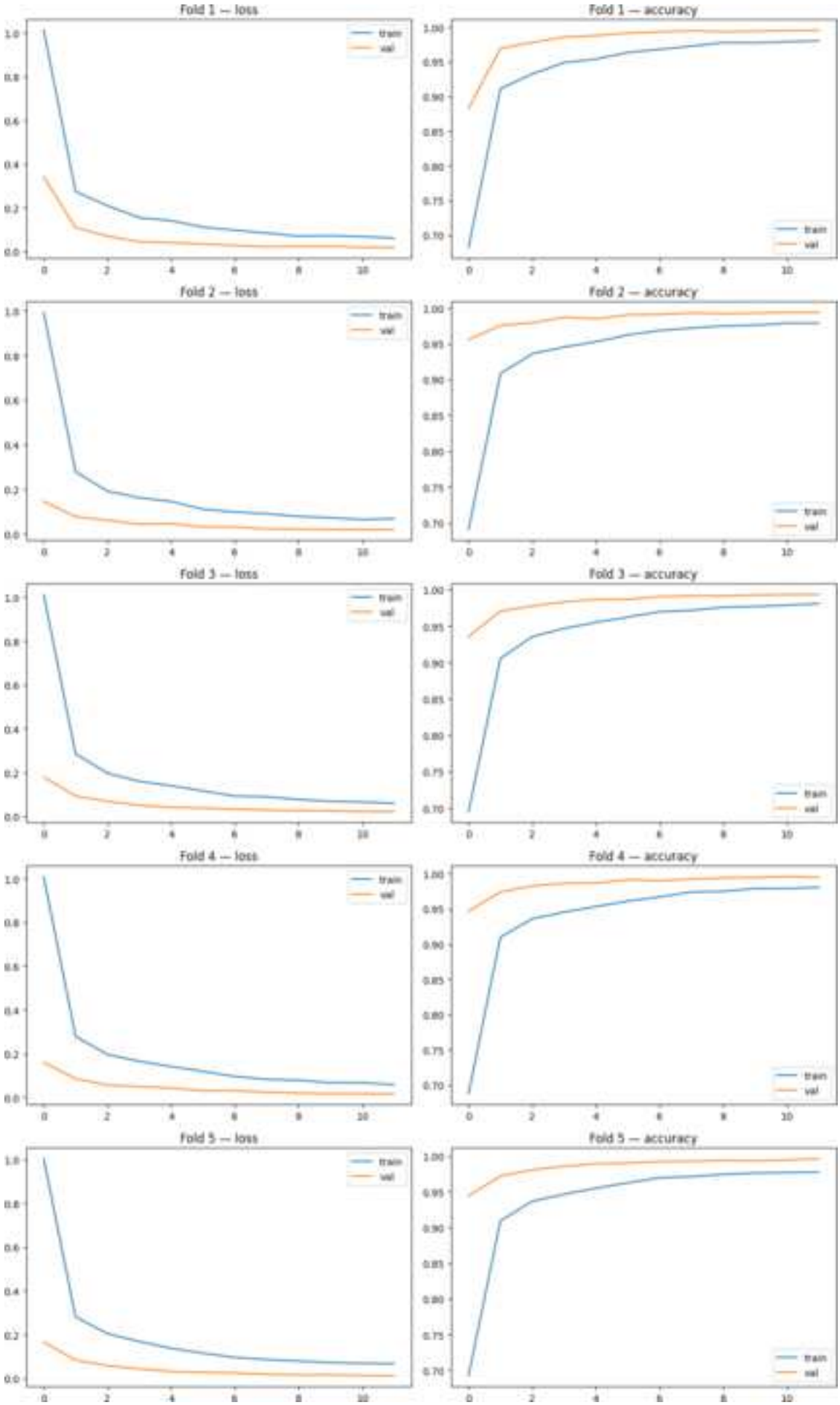
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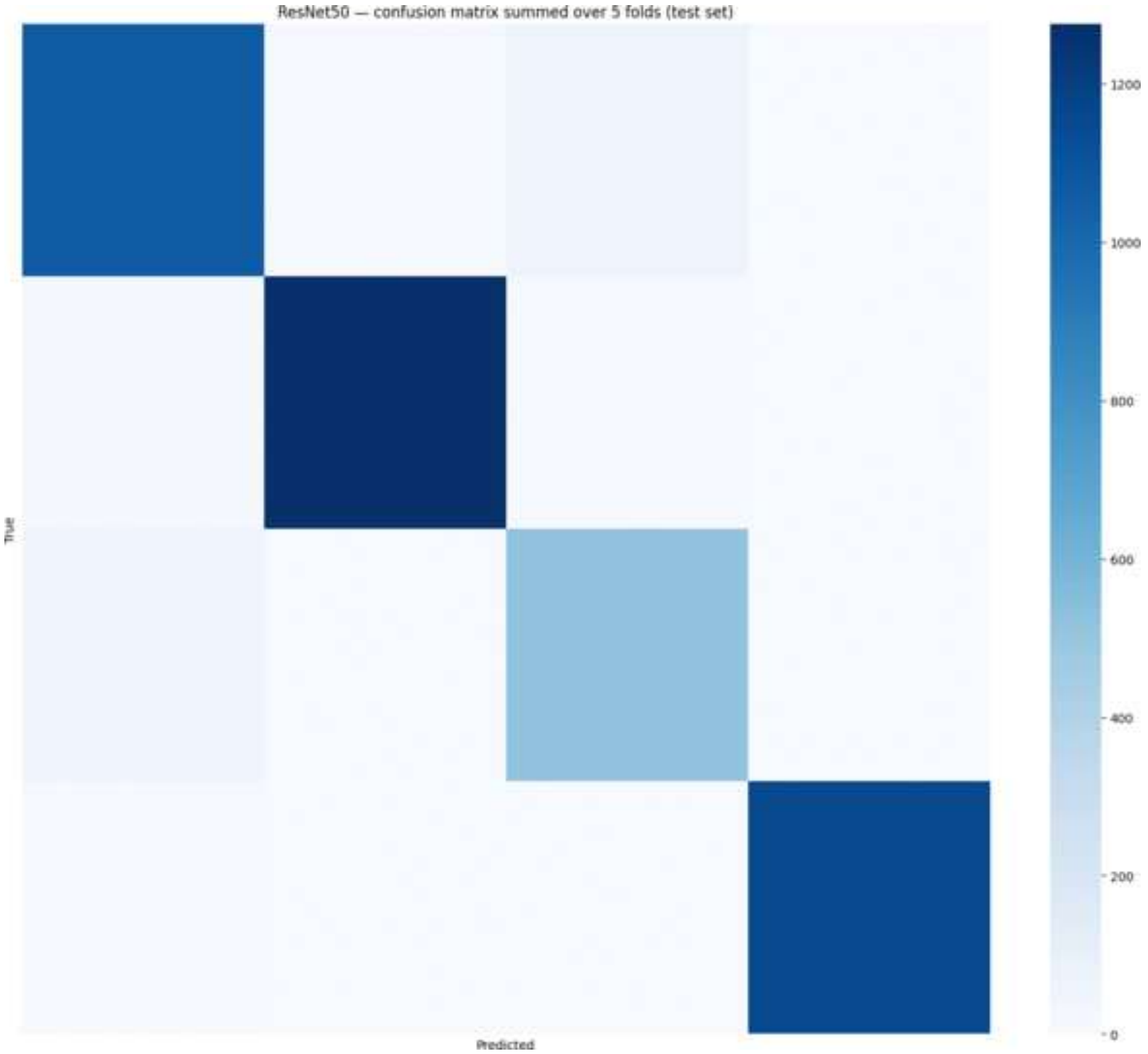
S5_Fig

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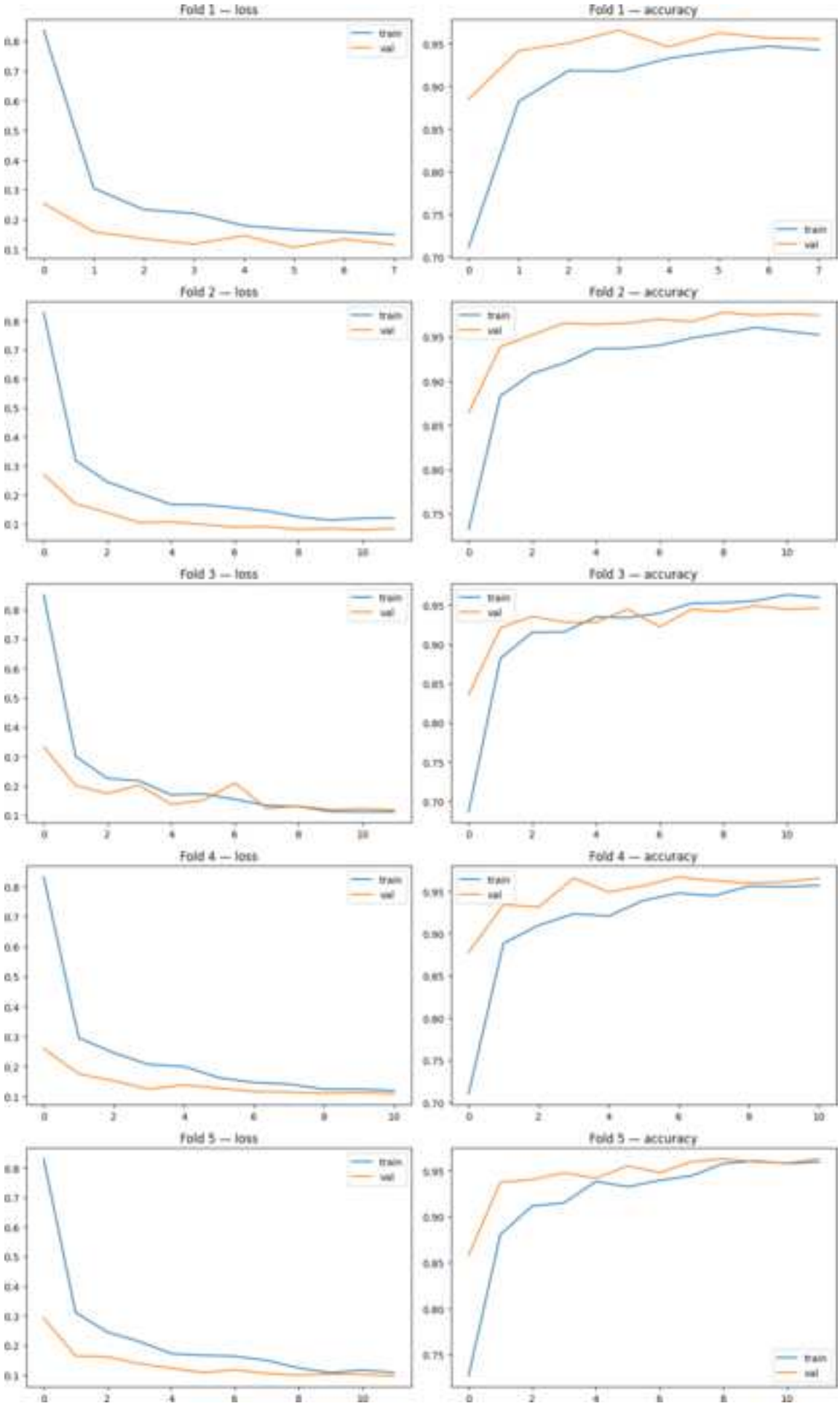
S6_Fig

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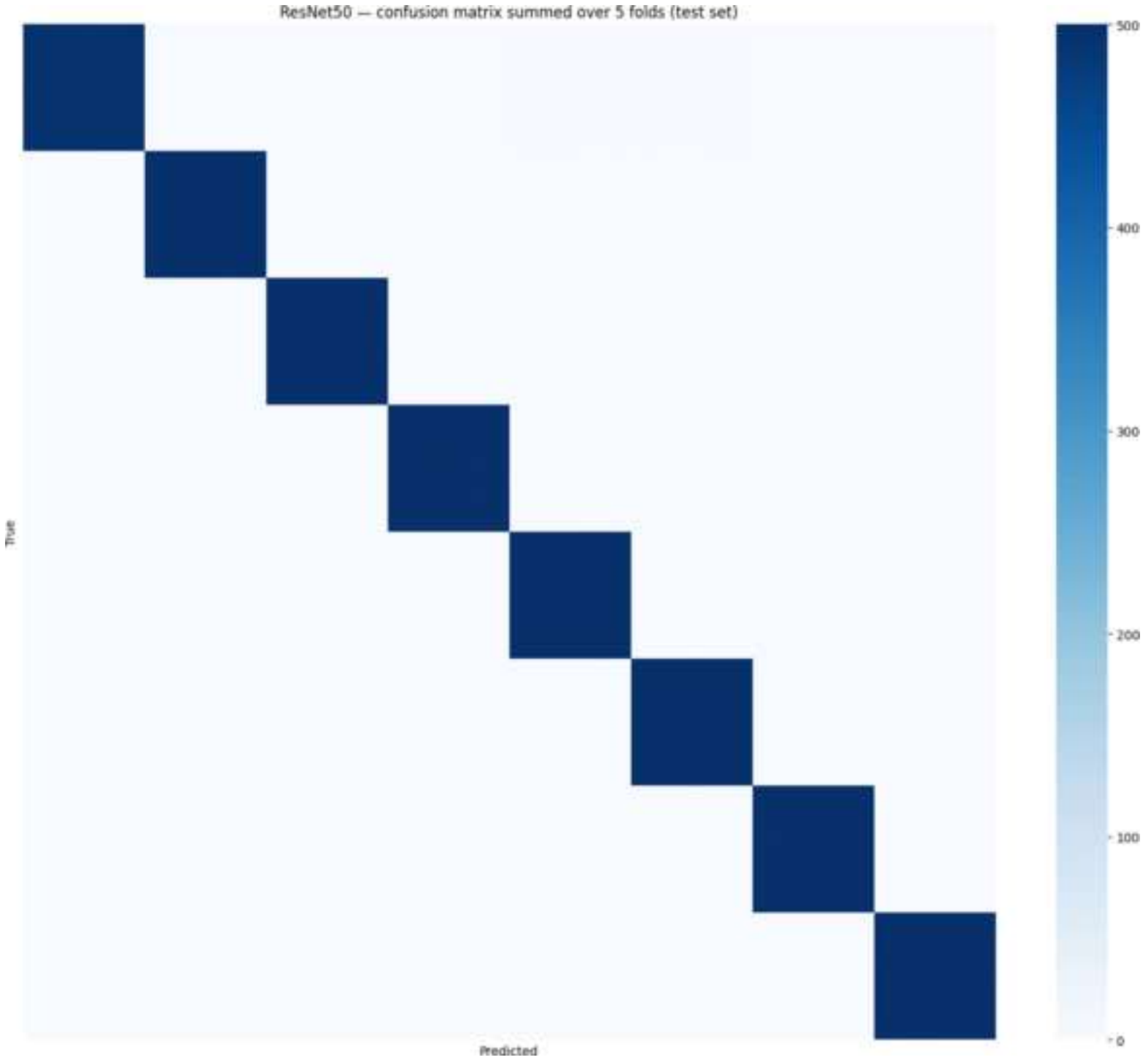
S7_Fig

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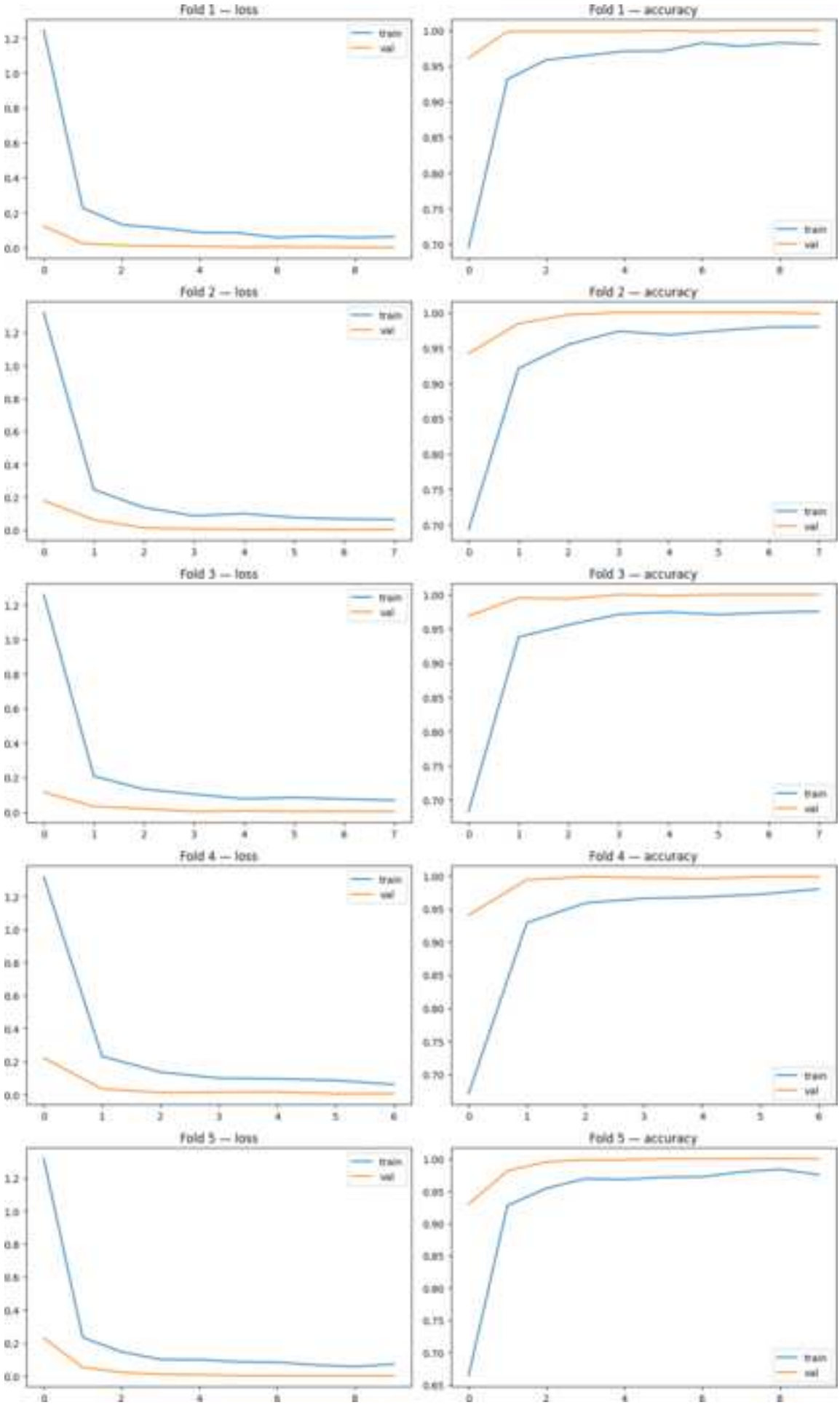
S8_Fig

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S9_Fig

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